Change Detection on Remote Sensing Images Using Dual-Branch Multilevel Intertemporal Network

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***Abstract*— Change detection (CD) of remote sensing (RS) images is mushrooming up accompanied by the on-going innovation of convolutional neural networks (CNNs). Yet with the high-speed technology upgrade, the obstacle that identifies unbalanced variations in foreground–background categories still lies on the table, especially in cases with limited samples and massive interference such as seasonal turnover, illumination intensity, and building reformation. Moreover, to date, neither of the off-the-shelf methods probes the feasibility of direct interaction between bitemporal images before accessing difference features. In this article, we propose a dual-branch multilevel intertemporal network (DMINet) to efficiently and effectively derive the change representations. Specifically, by unifying self-attention (SelfAtt) and cross-attention (CrossAtt) in a single module, we present an intertemporal joint-attention (JointAtt) block to steer the global feature distribution of each input, motivating information coupling between intralevel representations and meanwhile suppressing the task-irrelevant interferences. In addition, centering more on the detection of difference features, a reliable architecture is designed by spotlighting two concerns, i.e., the difference acquisition using subtraction and concatenation as well as the multilevel difference aggregation using incremental feature alignment. Based on a naive backbone without sophisticated structures, i.e., ResNet18, our model outperforms other state-of-the-art (SOTA) methods on four CD datasets, especially in cases with rarely samples. Moreover, the achievement is attained with light overheads.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), dual-branch difference acquisition, intertemporal joint-attention (JointAtt), multilevel aggregation, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) intends to monitor variations that occur in the same geographic region over multiple periods. With the sophistication and innovation of sensor devices, extensive multitemporal high-resolution RS images are available for land management [1], disaster assessment, and urban/rural development investigation [2], [3]. However, since multitemporal RS images are recorded at different imaging conditions and various photographing periods, massive disturbance of irrelevant changes often occurs, including seasonal turnover, illumination intensity, and building reformation.

Till now, most typical methods have been widely investigated, such as algebra-based methods, transformation-based methods, and classification-based methods. The algebra-based methods use arithmetic operations, such as image differencing [4], image ratioing [5], and image quantification [6], to derive change maps by generating difference features containing the change magnitudes. In contrast, image transformation methods, such as principal component analysis (PCA) [7], change vector analysis [8], and tasseled cap transformation [9], resort the spatial mapping to highlight the changed information. The classification-based methods which are represented by the traditional machine learning-based methods, such as support vector machines [10], K-nearest neighbors [11], and random forest regression [12], distinguish changed pixels by difference features or spatial–temporal features. Inevitably, the aforementioned approaches, as typical of unsupervised ones, suffer from extreme reliance on empirical design and yield unsatisfactory results on high-resolution RS images.

With technological innovations, deep-learning-based methods with impressive capabilities for harvesting nonlinear features have pushed traditional models aside. From early fully connected networks (FCNs) [13], [14], to typical convolutional neural networks (CNNs) [15], [16], recurrent neural networks (RNNs) [17], as well as graph convolutional networks (GCNs) [18], [19], [20], [21], and then to the recently rising star transformer [22], most deep-learning-based models embody the promising potential for application in CD tasks. For instance, in [23], the generalization capability of FCN is enhanced by allocating attention guidance. Incorporating certain clustering method to strengthen the discriminative characteristics of CNN is explored in [24] and [25]. To emphasize the importance of correlation between the feature pairs, Lee et al. present a local similarity Siamese network using cosine similarity measurement. For a similar purpose, Jiang et al. develop a pyramid-feature-based and attention-guided Siamese network. Qu et al. [18], Tang et al. [19], and Wang et al. [20] use GCN to flexibly capture the global characteristics and structure information. Besides, the multilevel Swin-transformer [26] and the U-shaped Swin-transformer [27] are proposed to exploit multihead attention straightforwardly for CD tasks.

Notably, the receptive fields of CNNs are incrementally expanded through network deepening, thereby allowing the acquisition of features ranging from subtlety to wholeness,

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while RNNs, GNNs, and transformers have remarkably robust capabilities for global context modeling. With regard to the absence of long-range dependencies in CNNs, certain attention strategies, such as channel-attention [28], spatial-attention [29], self-attention (SelfAtt) [30], [31], and cross-attention (CrossAtt) [32], [33], are introduced in pursuit for boosting the global representation capability. For instance, Li et al. [34] propose to use solo-attention for feature enhancement and co-attention for the exploration of feature associations between query–target sample pairs. Li et al. [30] propose a novel feature calibration module, which effectively enhances the discrimination of spatial information and reduces the redundancy. In [31], a bitemporal image transformer (BIT) is proposed to assemble long-range concepts from the features extracted by a naive CNN, whose detection performance and efficiency are satisfactory. However, in the encoding stage of BIT, the features of the bitemporal RS images are still aggregated by stacking them purely along the channel direction, which allows limited interaction between bitemporal features.

Overall, for all the mentioned methods, there exist some deficiencies that desire to be explored. First, CD puts more emphasis on the investigation of spatial–temporal changes, rather than squandering excessive energies on the, respectively, feature extraction of bitemporal images. Moreover, detecting targets with various scales under massive shooting-triggered interference involves a whole consideration of the reliability, robustness, and potential for the acquisition of multilevel differences. Accordingly, an efficient and effective model that consistently revolves around difference capturing is desired. Second, due to the uncertainty of change distribution in bitemporal images and the requirement to suppress the task-irrelevant interference such as seasonal turnover and building reformation, performing feature interaction before the difference acquisition plays an important role, which motivates and supports the counterpart to glimpse the representation of the other. In other words, feature interaction between bitemporal images facilitates both the parties to propagate their respective spatial information to mutually understand whether task-relevant targets are available. Third, considering the unbalanced dilemma between foreground–background categories, especially the presence of task-irrelevant yet shared objects, the attention should be concentrated on the changed regions with mutual guidance during interaction. As a mimic to the human behavior "self-introduction," propagating partial representations of bitemporal images to each other behaves as steering the attention distribution while preserving the own characteristics, which seems to be a sound alternative.

To relieve the addressed issues, we propose a dual-branch multilevel intertemporal network (DMINet) with a simple yet effective architecture. Specifically, we first, respectively, extract the multilevel features of the bitemporal images using a pretrained ResNet18. Then, aiming at fulfilling feature interactions between bitemporal images, the intertemporal joint-attention (JointAtt) module is introduced to mutually steer the attention distribution, acting on features with level shared and temporal differed and concentrating on changed regions under massive irrelevant interference. In addition, a dual branch of pixelwise subtraction and channelwise concatenation is designed to capture the potentially multilevel differences. Finally, a multilevel incremental aggregation module based on simple Hadamard products serves as a decoder before acquiring the prediction maps, which leverages multilevel features to satisfy the requirements of scale-varied change targets. The main contributions of our work are listed as follows.

1) A plug-and-play intertemporal JointAtt module is proposed to suppress task-irrelevant interference while steering attention to the genuine changes. Specifically, by incorporating both the self- and the cross-attentions in a single module, individual representations of bitemporal images are bonded, and global attention distribution bridging the information interaction is derived.
2) To spotlight the potential of difference acquisition, we design an efficient and effective CD architecture from two perspectives, including difference generation and multilevel aggregation. The former involves a dual branch of pixelwise subtraction and channelwise concatenated learning, while the latter performs incremental alignment via simple Hadamard product between multilevel differences.
3) Quantitative and qualitative experiments on four datasets demonstrate that our proposed DMINet consistently outperforms other state-of-the-art (SOTA) methods, especially in cases with limited samples available. The trained models and source code have been uploaded at https://github.com/ZhengJianwei2/DMINet.

The rest of this article is organized as follows. In Section II, a brief review of related work is presented. The specifics of our proposed framework are described in Section III. A comprehensive experimental evaluation is conducted in Section IV. Finally, Section V concludes the article.

II. Related Work

According to the characteristic of bitemporal images in the CD task, the existing deep-learning-based approaches can be generally divided into two main streams. One is the image-level approach that straightforwardly combines the inputs and feeds them into a single segmentation network for detection. Another is the feature-level option that derives the difference features by certain operations, i.e., subtraction, multiplication, and concatenation, on the high-level features extracted from the backbone, which is advocated to be more effective. In the sequel, we roughly review the feature extraction and information fusion strategies of two branches, i.e., CNN-based and transformer-based models, which are both the rooted studies of our work.

A. CNN-Based Methods

To cope with irrelevant changes and sophisticated objects, considerable recent researches intend to enhance the generalization ability of CNNs by incorporating strategies that can facilitate contextual information modeling, which is crucial for distinguishing changes from bitemporal RS images. Among

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them, the multilevel feature fusion strategy [35], [36], [37] is second to none, which combines low-level features that facilitate target localization and high-level features with sufficient semantic information. For instance, Zhang et al. [35] impose constraints explicitly on the difference feature maps recorded by various stages of the encoder for yielding finer representations. Li et al. [38] propose a densely attentive refinement network based on the U-shaped encoder–decoder architecture, which supports the progressive adjustment of the predicted change maps at each stage of the decoding process. Moreover, several schemes are extensively investigated and adopted to expand the scope of the receptive field, including deepened network architectures [37], dilated convolutional strategies [39], and attention mechanisms [35], [36], [37], [40]. Notably, attention mechanisms have evolved in various forms, such as channel attention [37] that can recalibrate channelwise feature responses and spatial attention [40] with the ability to highlight task-relevant spatial features.

Note again that the multilevel feature maps with local characteristics derived by CNNs are pivotal for modeling spatial–temporal contextual information in the CD task. However, CNNs lack the ability to model long-range dependencies, leaving room for enhancement. Besides, neither of the available CD methods addresses the direct interaction between bitemporal images before acquiring difference features. Therefore, in this article, we introduce intertemporal JointAtt modules for feature coupling and information communication after feature extraction by a naive CNN, which enables individuals to glimpse partial representations of each other while preserving their respective features.

B. Transformer-Based Methods

Recently, a storm involving the field of computer vision has been set off by the emergence of transformers. With the powerful and robust representational capability on large-scale datasets, the transformer-based models demonstrate equivalent or even superior performance as the convolutional counterparts, including image classification [22], semantic segmentation [22], [41], object detection [22], [42], image generation [43], video retrieval [44], and target tracking [45].

At the same time, the CD tasks cannot escape the fate of being infiltrated by transformer. Song et al. [26] apply multiple Swin-transformer blocks to aggregate the multilevel feature maps of each image captured by the CNN, and then calculate the difference features between the bitemporal images. Yet, this approach is undoubtedly extremely resource-consuming in terms of memory and computation. Similarly, Zhang et al. [27] construct a pure Swin-transformer with a Siamese U-shaped structure for the CD task, which uses utmost exaggerated 32 Swin-transformer blocks. In the recent research [46], an intrascale cross-interaction and interscale feature fusion network (ICIF-net) is proposed to tap the potential of integrating CNN and transformer in a parallelized manner. The intrascale cross-interaction module revolves around the complementarity between fine-grained local details and target-level global interests, while the interscale feature fusion module is intended to integrate high-level representations that benefit localization and low-level features that facilitate structural sketching. Although ICIF-Net achieves significant detection results, the incorporation of the entire transformer backbone occupies a great number of parameters. As the exploration of transformer in CD, the aforementioned methods concentrate mainly on detection performance without regarding the efficiency. In contrast to developing a sophisticated architecture, Chen et al. [31] propose a BIT to efficiently and effectively model the spatial–temporal contexts, which lightweightly consists of a transformer encoder and two transformer decoders. However, the architecture of BIT might be further optimized, such as the transformer encoder can be simplified to a single-head attention, through which feature coupling and information communication is clearly more efficient, and the transformer decoders can be substituted with lightweight modules.

III. Methodology

A. Architecture Overview

The architecture of the proposed DMINet is illustrated in Fig. 1, which delivers an attractive and efficient solution for the CD task on RS images by scheduling the early feature interactions between bitemporal images. Specifically, the bitemporal RS images are first extracted by the two parallel branches of Siamese pretrained CNN (ResNet18). In our implementation, multilevel feature maps with spatial sizes of 64×64, 32×32, and 16×16 at the first three stages of ResNet18 are taken. Next, the intralevel bitemporal features are encoded interactively with a lightweight encoder, i.e., the intertemporal JointAtt module, in the manner of mutually guiding the attention distribution of each other. This early interaction between bitemporal features not only enables the model to suppress irrelevant interference caused by various imaging conditions but also allows the model attention to be focused on the genuinely changed regions. Afterward, the coupled multilevel bitemporal features are directly converted into difference maps in a dual branch of subtraction and concatenation before being transmitted into the multilevel feature alignment module. Notably, the pixel-level subtraction operation approximates the difference pattern with the absence of extremely valuable information in the bitemporal feature maps, while the concatenation operation approaches the changes gradually in a learnable manner. Finally, the two aligned difference maps are coded by conventional prediction heads and then combined with the argmax operation to generate the prediction map. As a whole, our DMINet emphasizes more on the intertemporal interaction between bitemporal images and the acquisition of precise difference features, which are crucial for CD mission.

B. Intertemporal JointAtt Module

Note again that due to multitemporal photographing, the captured images suffer severely from task-irrelevant interference such as seasonal turnover, illumination diversity, and building reformation. This issue is blamed as one of the main causes of the semantic gap between bitemporal features. As a result, the investigation of intertemporal feature interactions is required in advance of capturing the genuine differences.

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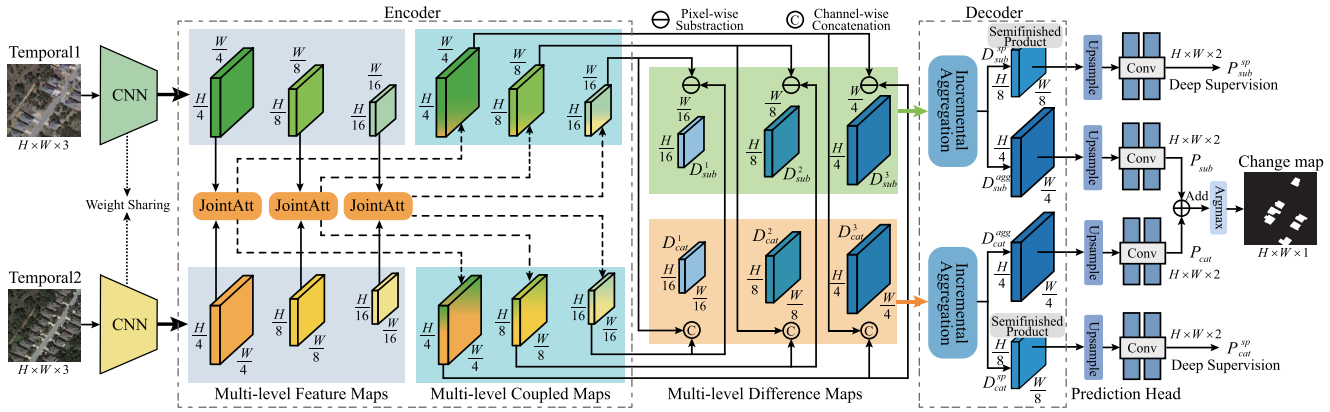


Fig. 1. Architecture of the proposed DMINet. The intralevel feature maps of bitemporal images extracted by pretrained CNN are preliminarily coupled through intertemporal JointAtt, followed by two strategies of pixelwise subtraction and channelwise concatenation to acquire the change differences. Then, feature alignment modules are constructed for multilevel difference aggregation. Finally, the dual-branch predictions are derived by conventional prediction heads consisting of upsampling and convolutions, followed by addition and argmax operations to generate the final map.

Moreover, considering that numerous task-irrelevant objects are shared in the bitemporal images, the changed targets would probably account for a minority of regions, especially in the case of unbalanced foreground–background categories. To tackle these challenges, we intend to facilitate the attention distribution on the genuinely changed regions by mutually steering between the bitemporal images. In the implementation, both the SelfAtt and CrossAtt are jointly considered in a bonded module.

The features are first mapped via a 1×1 convolutional layer and reshaped into embeddings consisting of query Q, key K, and value V. Technically, the conventional SelfAtt mechanism, as formulated in (1), behaves as mapping a query (Q_i) and a set of key–value ($K_i - V_i$) pairs to a weighted sum of V_i, where the weight assigned to each V_i is the similarity between Q_i and the corresponding K_i

$$\begin{aligned} \text{SelfAtt}(Q_i, K_i, V_i) &= \text{Softmax}\left(Q_i \cdot K_i^T\right) \cdot V_i \\ &= A_i^{\text{Self}} \cdot V_i. \end{aligned} \tag{1}$$

In (1), the vector sequences Q_i, K_i, $V_i \in R^{N \times C} (C \ll N)$ with $N = H \times W$, H, and W are, respectively, the height and weight of the features, C denotes the channel dimension, Softmax is the activation function, T represents a transpose operation, and A_i^{Self} is the SelfAtt distribution. Nevertheless, SelfAtt merely spotlights interests in the global space of a single image, regardless of the interaction between temporals, which cannot reach our expectations. Acting on the coupling between a pair of images, as formulated in (2), the CrossAtt that swaps the query Q based on SelfAtt may seem like a promising solution

$$\begin{aligned} \text{CrossAtt}(Q_j, K_i, V_i) &= \text{Softmax}\left(Q_j \cdot K_i^T\right) \cdot V_i \\ &= A_{j,i}^{\text{Cross}} \cdot V_i \end{aligned} \tag{2}$$

where i and j denote the serial numbers of different images ($j \neq i$), and A_i^{Cross} is the CrossAtt distribution. CrossAtt entirely uses Q of the counterpart to learn the global attention distribution with its own K, which may cause certain information loss. Therefore, we attempt to combine Qs of bitemporal features and achieve the global attention distribution with complementary Ks, benefitting from both SelfAtt and CrossAtt.

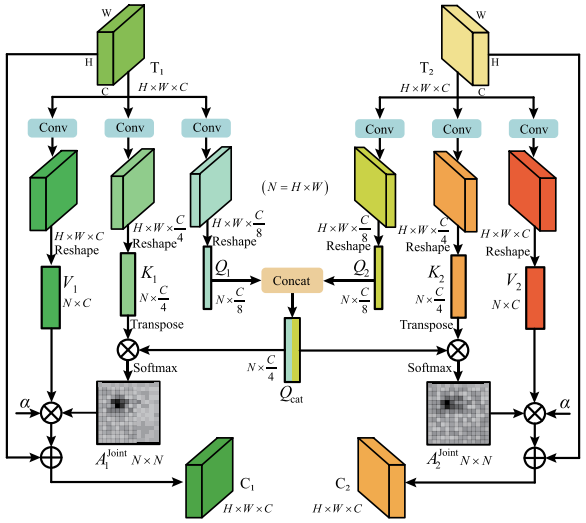


Fig. 2. Illustration of the intertemporal JointAtt module.

As illustrated in Fig. 2, an intertemporal interaction that incorporates partial representations of the opposite side while retaining all the personal information is used. Taking the JointAtt distribution $A_1^{\text{Joint}} \in R^{N \times N}$ attained from the first-temporal image as an example, with Q_1 and Q_2 concatenated into Q_{cat}, the similarity between Q_{cat} and K_1 can then be derived by a sequential treatment of multiplication and Softmax. Since Q_{cat} involves the complementary queries from bitemporal images, both SelfAtt and CrossAtt can be learned simultaneously. The flowchart is given in Fig. 3. Mathematically, the overall intertemporal JointAtt can be formulated in the following:

$$\begin{aligned} C_i &= \text{JointAtt}(\{Q_i, Q_j\}, \{K_i, V_i\}) \\ &= \alpha \cdot \text{Softmax}\left(\text{Concat}(Q_1, Q_2) \cdot K_i^T\right) \cdot V_i + T_i \\ &= \alpha \cdot \text{Joint}\left(A_i^{\text{Self}}, A_{i,j}^{\text{Cross}}\right) \cdot V_i + T_i \\ &= \alpha \cdot A_i^{\text{Joint}} \cdot V_i + T_i \end{aligned} \tag{3}$$

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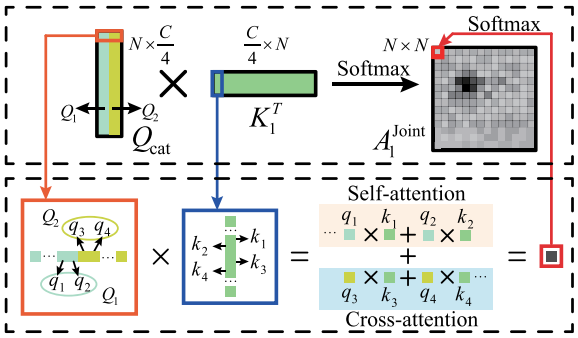


Fig. 3. Illustration of calculating JointAtt distribution.

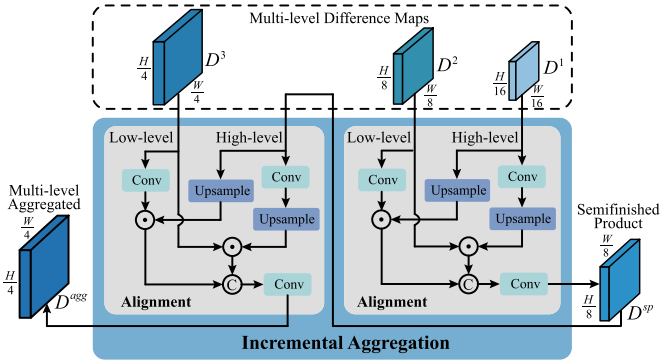


Fig. 4. Illustration of the multilevel incremental aggregation.

where the shared parameter α is initialized to 1 and then participates in back propagation for adaptive adjustment, Joint(A_i^{Self}, $A_{i,j}^{\text{Cross}}$) indicates that the distribution A_i^{Joint} involves both SelfAtt and CrossAtt, and the raw input T_i is attached by a skip connection. Functionally, the interaction motivates and supports an individual image to glimpse the representation of another RS input while preserving its own features, which efficiently bridges the information conduction and exploits the spatial–temporal contextual characteristics. In addition, the simultaneous actions of the self-control and the guidance from counterpart hold the potential of suppressing the task-irrelevant interference to some extent, which further promotes the learning of global attention centering more on the genuinely changed regions.

C. Dual-Branch Difference Acquisition

With the respective information of different temporal images exchanged through the intertemporal JointAtt module, the remaining nuisance that requires an urgent solution is how to attain the difference features between inputs. Confronted with the extreme imbalance between two categories of the foreground and background in CD assignment, it is essential to maximize the potential of the acquisition method while guaranteeing its stability and reliability.

As is known, common treatments include pixelwise operations such as subtraction, multiplication, or parameter-dependent learning. Yet for CD, using any of these operations alone is suboptimal. How to derive the more elaborate difference information deserves a deeper investigating. On one hand, the difference intuitively captured by a simple subtraction operation is surely sensitive to task-irrelevant changes such as seasonal turnover and building reformation. On the other hand, while concatenated learning hammers at circumventing the above-mentioned shortcomings, it counts heavily on massive training samples and is probably unpredictable and unexplainable. Therefore, as illustrated in Fig. 1, we model the structure of dual branch simultaneously using pixelwise subtraction and channelwise concatenation, which can suppress the task-irrelevant disturbance while facilitating the optimization toward a credible and promising orientation.

D. Multilevel Difference Alignment

It is crucial for the CD task that multilevel outcomes at different stages of the naive CNN cover various levels of detection targets. On one hand, high-resolution and low-level features are equipped with considerable fine-grained details such as target boundaries and spatial structures, yet their sensitivity to background noise is deplorable. On the other hand, low-resolution and high-level representations convey more abstract semantic features, which are instrumental for region localization and noise suppression.

Refraining from elaborate architecture practices of feature decoders, we fulfill a lightweight aggregation of features from high to low levels and in an incremental fashion. Specifically, since the values of changed regions on difference maps tend to be larger, we adopt a pixelwise multiplication operation to further highlight the features. In other words, the Hadamard product is performed between a lower level feature and an upsampled higher level feature, which evolves to highlight the task-relevant features. The process is illustrated in Fig. 4. Taking multilevel difference maps D^2 and D^1 as examples, the multilevel feature incremental aggregation can be described as follows:

$$D^{\text{sp}} = \text{Conv}(\text{Concat}(\text{Up}(\text{Conv}(D^1)) \circ D^2, \text{Conv}(D^2) \circ \text{Up}(D^1))) \quad (4)$$

where $\circ$, Conv, Up, and Concat represent the operations of the Hadamard product, convolution, upsampling, and merging along channels, respectively. D^{sp} is the semifinished product of the proposed multilevel feature aggregation, which is equipped with deep supervision in the optimization process. Moreover, given D^3 and D^{sp}, the outcome D^{agg} aggregating all the three-level difference features can be written as

$$D^{\text{agg}} = \text{Conv}(\text{Concat}(\text{Up}(\text{Conv}(D^{\text{sp}})) \circ D^3, \text{Conv}(D^3) \circ \text{Up}(D^{\text{sp}}))). \quad (5)$$

By gradually aligning the difference maps with different levels, the model is capable of facilitating an accurate localization and capturing the fine details for various scales of targets, especially for extremely tiny changes.

E. Prediction Head and Loss Function

In the prediction phase, as shown in Fig. 1, the conventional prediction heads used in BIT [31] are incorporated. Specifically, the aggregated maps are upsampled to the size of the raw inputs by bilinear interpolation, followed by a conversion into difference predictions by two convolutional layers and a batch-normalization operation. Owing to the dual-branch

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architecture of difference acquisition and multilevel incremental aggregation, we generate altogether two predictions P_{sub}, $P_{\text{cat}} \in R^{H\times W\times 2}$ and two semi-products $P_{\text{sub}}^{\text{sp}}$, $P_{\text{cat}}^{\text{sp}} \in R^{H\times W\times 2}$, in which the two channels denote the foreground and background, respectively. In the training phase, we optimize the model parameters by minimizing the cross-entropy loss L_{ce} between four predictions and the ground truth (GT) one by one

$$L_{\text{ce}}(Y, P) = -\frac{1}{N}\sum_{i=1}^{N}[Y_i \log_2 P_i + (1 - Y_i)\log_2(1 - P_i)] \tag{6}$$

where N is the total number of pixels in a change map. The total training loss is formulated as

$$\begin{aligned} L_{\text{total}} = {} & \lambda_1 L_{\text{ce}}(\text{GT}, \sigma(P_{\text{sub}})) + \lambda_2 L_{\text{ce}}(\text{GT}, \sigma(P_{\text{cat}})) \\ & + \lambda_3 L_{\text{ce}}\left(\text{GT}, \sigma\left(P_{\text{sub}}^{\text{sp}}\right)\right) + \lambda_4 L_{\text{ce}}\left(\text{GT}, \sigma\left(P_{\text{cat}}^{\text{sp}}\right)\right) \end{aligned} \tag{7}$$

where σ is the sigmoid function, and λ_1, λ_2, λ_3, and λ_4 are the tradeoff parameters applied to regulate the influence of each loss. In practice, we consider that the two predictions (resp. semi-products), i.e., P_{sub} and P_{cat} (resp. $P_{\text{sub}}^{\text{sp}}$ and $P_{\text{cat}}^{\text{sp}}$), contribute equally to the overall loss. Therefore, to ease model selection, $\lambda_1 = \lambda_2$ and $\lambda_3 = \lambda_4$ are performed in the experiments. Note that $P_{\text{sub}}^{\text{sp}}$ and $P_{\text{cat}}^{\text{sp}}$ are imposed only as depth supervisors in the loss function, facilitating a more accurate change map for P_{sub} and P_{cat}. In the inference stage, we first pixelwisely perform the addition operation between P_{sub} and P_{cat}. Then, an argmax operation is appended to convert the two-channel map into a single-channel map. The final values contain 1 and 0, denoting whether the pixel is changed or not.

IV. Experiments

A. Data Description

Extensive experiments are conducted on four representative CD datasets to verify the practical performance of the proposed DMINet.

1) LEarning, VIsion, and Remote sensing (LEVIR-CD) [50] is a large-scale and very-high-resolution (VHR) CD dataset, which contains 637 pairs of RS images with a time span of 5–14 years, a spatial size of 1024 × 1024, and a spatial resolution of 0.5 m/pixel. These images were derived from 20 various areas in multiple cities in Texas, USA, covering 31 333 changed building instances. Notably, LEVIR-CD covers various types of buildings with significant land-use variations, such as villa residences, tall apartments, small garages, and large warehouses. Due to the limited GPU memory capacity, we cropped each image into 16 nonoverlapping patches with a size of 256 × 256. In this way, LEVIR-CD applies 7120/1024/2048 pairs of patches for training/verification/testing, respectively.
2) Wuhan University (WHU-CD) [51] records the construction of buildings in the region of Christchurch, New Zealand, after a magnitude 6.3 earthquake in February 2011. Specifically, the dataset contains a pair of aerial images with a spatial size of 32 507 × 15 354 and a resolution of 0.2 m/pixel taken over the same area in 2012 and 2016. Since the solution of how to split the samples is not given in [51], we cut the large-scale image pairs into 256 × 256 patch pairs and randomly divide them into 6096/762/762 for sets of training/verification/test.
3) Guangzhou Dataset (GZ-CD) [52] is a VHR satellite image CD dataset with a spatial resolution of 0.55 m/pixel, which covers the suburb areas of Guangzhou City, China, from 2006 to 2019. Through the Google Earth service, a total of 19 season-varying image pairs with spatial size ranging from 1006 × 1168 to 4936 × 5224 are obtained. In consistent with the processing of the previous two datasets, we uniformly cut the inputs into the size of 256 × 256. These image patches are further divided into 2504/313/313 pairs for training/verification/test sets, respectively, which implies that GZ-CD is a comparatively small-sized dataset compared with LEVIR-CD and WHU-CD.
4) Sun Yat-Sen University Dataset (SYSU-CD) [36] includes 20 000 orthographic aerial pairs with a spatial size of 256 × 256 and a spatial resolution of 0.5 m, which are gathered in Hong Kong. Noteworthy, the SYSU-CD dataset covers various types of targets in addition to buildings, such as vessels, roads, and vegetation, which presents a tougher ordeal for the CD tasks. For training/validation/testing sets, we allocate 8000/4000/8000 samples, respectively.

B. Experimental Setup

1) Comparison to SOTAs: To authenticate the effectiveness and efficiency of our DMINet on bitemporal RS images, several SOTA models are picked as the competitors, including FC-EF [47], FC-Siam-Di [47], FC-Siam-Conc [47], FresUNet [48], image fusion network (IFNet) [35], dual-task constrained deep siamese convolutional network (DTCDSCN) [40], densely connected siamese network (SNUNet) [37], BIT [31], edge-guided recurrent CNN (EGR-CNN) [17], deep multiscale siamese network with parallel convolutional structure and self-attention (MSPSNet) [49], and ICIFNet [46]. The specific properties held by different methods are presented in Table I. From this table, we can observe that our DMINet enjoys all the favorable properties to guarantee an appealing performance, including the multilevel feature extraction and feature fusion, the difference acquisition using both concatenation and subtraction, and the direct usage of intertemporal integration. Comparatively, all the competitors lack consideration of some properties. Especially, the intertemporal integration module plays the role of learning global attention with acceptable computational cost, which is a critical and exclusive point of our model.

2) Implementation Details: In our experiments, the proposed DMINet is implemented using the PyTorch framework. In consistent with other comparison experiments, data enhancement, including flipping, rescaling, cropping, and Gaussian blurring, is performed in the training phase. Our model is trained with the stochastic gradient descent (SGD) algorithm using the batch size of 8, the initial learning rate (lr) of 1e-2, the momentum of 0.99, the weight decay of 5e-4,

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TABLE I

COMPARISONS OF MODEL EFFICIENCY AND ARCHITECTURE. THE PARAMS. AND FLOPS ARE GIVEN FOR EACH ALGORITHM EQUIPPED WITH DIFFERENT BACKBONES. THE USED IMAGES ARE OF SIZE 256 × 256 × 3

Models	Backbone	Params.(M)	FLOPs(G)	Fusion-level		Difference		Multi-level feature	Inter-temporal interaction
				Image	Feature	Concat	Subtract		
FC-EF [47]	UNet	1.35	3.57	✓		✓		✓	
FC-Siam-Di [47]	UNet	1.35	4.72		✓		✓	✓	
FC-Siam-Conc [47]	UNet	1.55	5.32		✓	✓		✓	
FresUNet [48]	UNet	1.11	2.02		✓	✓		✓	
IFNet [35]	VGG16	50.71	82.26		✓	✓	✓	✓	
DTCDSCN [40]	SE-ResNet34	41.07	20.44		✓	✓		✓	
SNUNet [37]	NestedUNet	12.04	54.82		✓	✓		✓	
BIT [31]	ResNet18	3.55	10.60		✓		✓		
EGRCNN [17]	UNet	9.64	17.64		✓		✓	✓	
MSPSNet [49]	-	2.21	14.17		✓	✓		✓	
ICIFNet [46]	ResNet18 & PVTv2-B1	23.82	25.36		✓	✓		✓	
Ours	ResNet18	6.24	14.42		✓	✓	✓	✓	✓

TABLE II

QUANTITATIVE COMPARISONS ON BUILDING CHANGES. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%)

Methods	LEVIR-CD Pre. / Rec. / F1 / IOU / OA / Kappa	WHU-CD Pre. / Rec. / F1 / IOU / OA / Kappa	GZ-CD Pre. / Rec. / F1 / IOU / OA / Kappa
$FC\text{-}EF_{18}$	85.56 / 78.76 / 82.02 / 69.52 / 98.24 / 81.40	78.79 / 68.15 / 73.08 / 57.58 / 97.61 / 72.62	81.29 / 68.61 / 74.41 / 59.25 / 95.74 / 72.34
$FC\text{-}Siam\text{-}Di_{18}$	91.53 / 78.24 / 84.36 / 72.95 / 98.52 / 83.71	80.44 / 54.59 / 65.04 / 48.19 / 97.20 / 63.67	80.36 / 61.80 / 69.87 / 53.69 / 95.19 / 67.36
$FC\text{-}Siam\text{-}Conc_{18}$	84.17 / 79.49 / 81.77 / 69.16 / 98.19 / 80.91	75.89 / 69.30 / 72.61 / 56.99 / 97.51 / 71.74	80.37 / 68.97 / 74.23 / 59.03 / 95.68 / 72.09
$FresUNet_{19}$	89.86 / 86.41 / 89.05 / 80.26 / 98.92 / 88.47	86.55 / 77.68 / 81.88 / 69.32 / 98.36 / 81.09	87.97 / 77.44 / 82.48 / 70.65 / 97.10 / 80.82
$IFNet_{20}$	**94.39** / 82.42 / 88.01 / 78.58 / 98.86 / 87.43	92.24 / 75.78 / 83.11 / 71.12 / 98.59 / 82.62	87.39 / 64.57 / 77.65 / 63.47 / 96.64 / 76.08
$DTCDSCN_{21}$	89.25 / 85.68 / 87.43 / 77.67 / 98.75 / 87.16	78.65 / 81.24 / 79.92 / 66.56 / 98.05 / 78.68	86.08 / 73.86 / 79.51 / 66.01 / 96.56 / 77.95
$SNUNet_{22}$	89.18 / 87.17 / 88.16 / 78.83 / 98.82 / 87.65	85.58 / 80.99 / 83.22 / 71.26 / 98.44 / 82.81	85.81 / 79.63 / 83.32 / 71.88 / 97.19 / 81.61
BIT_{22}	89.24 / 89.37 / 89.31 / 80.68 / 98.92 / 88.86	86.64 / 81.39 / 83.98 / 72.39 / 98.52 / 83.74	88.23 / 79.23 / 83.88 / 72.24 / 97.27 / 82.31
$EGRCNN_{22}$	92.83 / 86.46 / 89.59 / 81.12 / 98.92 / 88.92	90.23 / 81.09 / 84.86 / 75.03 / 98.67 / 84.07	87.91 / 80.36 / 83.96 / 72.36 / 97.23 / 81.97
$MSPSNet_{22}$	91.38 / 87.08 / 89.18 / 81.36 / 99.03 / 88.49	89.95 / **86.70** / 88.29 / 79.04 / 98.90 / 87.73	86.97 / 81.89 / 84.35 / 72.94 / 97.26 / 83.04
$ICIFNet_{22}$	91.32 / 88.64 / 89.96 / 81.75 / 98.99 / 89.44	92.98 / 85.56 / 88.32 / 79.24 / 98.96 / 87.75	**90.17** / 81.57 / 85.65 / 74.91 / 97.53 / 84.23
Ours	92.52 / **89.95** / **90.71** / **82.99** / **99.07** / **90.19**	**93.84** / 86.25 / **88.69** / **79.68** / **98.97** / **87.89**	89.31 / **83.90** / **86.52** / **76.24** / **97.64** / **85.23**

TABLE III

QUANTITATIVE COMPARISONS ON MULTIPLE CHANGE TYPES

Methods	SYSU-CD Pre. / Rec. / F1 / Iou / OA / Kappa
FC-EF [47]	72.28 / 72.85 / 72.57 / 56.94 / 87.01 / 66.26
FC-Siam-Di [47]	84.94 / 51.45 / 64.08 / 47.15 / 86.40 / 73.17
FC-Siam-Conc [47]	83.03 / 71.62 / 76.89 / 62.45 / 89.35 / 71.05
FresUNet [48]	78.56 / 74.96 / 76.38 / 61.85 / 89.12 / 69.54
IFNet [35]	79.59 / 73.58 / 76.53 / 61.91 / 89.17 / 69.68
DTCDSCN [40]	83.19 / 77.25 / 80.11 / 66.82 / 90.96 / 73.95
SNUNet [37]	83.49 / 76.37 / 79.77 / 66.35 / 90.87 / 73.69
BIT [31]	78.73 / 75.68 / 77.17 / 62.83 / 89.44 / 70.31
EGRCNN [17]	84.36 / 78.41 / 81.64 / 69.02 / 91.64 / 76.34
MSPSNet [49]	78.47 / 74.91 / 77.39 / 63.13 / 89.43 / 69.94
ICIFNet [46]	83.37 / 78.51 / 80.74 / 68.12 / 91.24 / 74.59
Ours	**85.03** / **79.86** / **82.08** / **69.60** / **91.67** / **76.62**

and the training epochs (max_epochs) of 200. Specifically, the learning rate decays linearly with epochs, whose formulation is shown in the following:

$$\text{deacy_lr} = \text{lr} * \left(1.0 - \frac{\text{current_epoch}}{\text{max_epochs} + 1}\right) \tag{8}$$

where current_epoch represents the current epoch number, and deacy_lr is the learning rate for current epoch. For fairness, all the experiments are consistently conducted on the workstation with Intel Xeon CPU E5-2690 v4 cores and two GPUs of NVIDIA TITAN V with a single 12G of video memory.

3) Evaluation Metrics: For the performance measurement of similarity between the predicted change probability maps and the GT, we introduce six metrics, including precision (Pre.), recall (Rec.), F1-score (F1), intersection over union (IOU), overall accuracy (OA), and Kappa. The metrics can be individually defined as follows:

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{9}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{10}$$

$$\text{F1} = 2/(\text{Recall}^{-1} + \text{Precision}^{-1}) \tag{11}$$

$$\text{IOU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{12}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{13}$$

$$\text{Kappa} = (\text{OA} - P)/(1 - P) \tag{14}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative, respectively. P in Kappa denotes the hypothetical probability of the chance agreement between reference and prediction, which can be formulated as

$$P = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{FN} + \text{TN})(\text{TP} + \text{TN})}{(\text{TP} + \text{FP} + \text{TN} + \text{FN})^2}. \tag{15}$$

Typically, a high precision (resp. recall) indicates a low commission (resp. omission) error. The remaining four metrics, i.e., F1, IOU, OA, and Kappa, are indicative of general model behavior, for which high values account for high detection accuracy.

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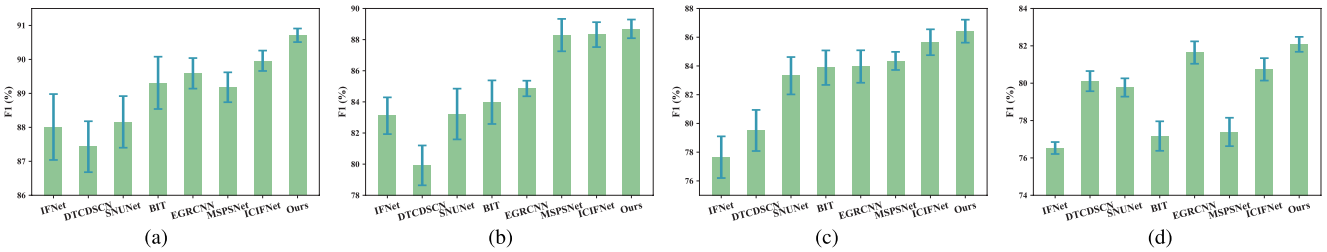


Fig. 5. Comparisons on four datasets. The bars indicate the F1 values, with the standard deviation shown at the top. (a) Methods on LEVIR. (b) Methods on WHU. (c) Methods on GZ. (d) Methods on SYSU.

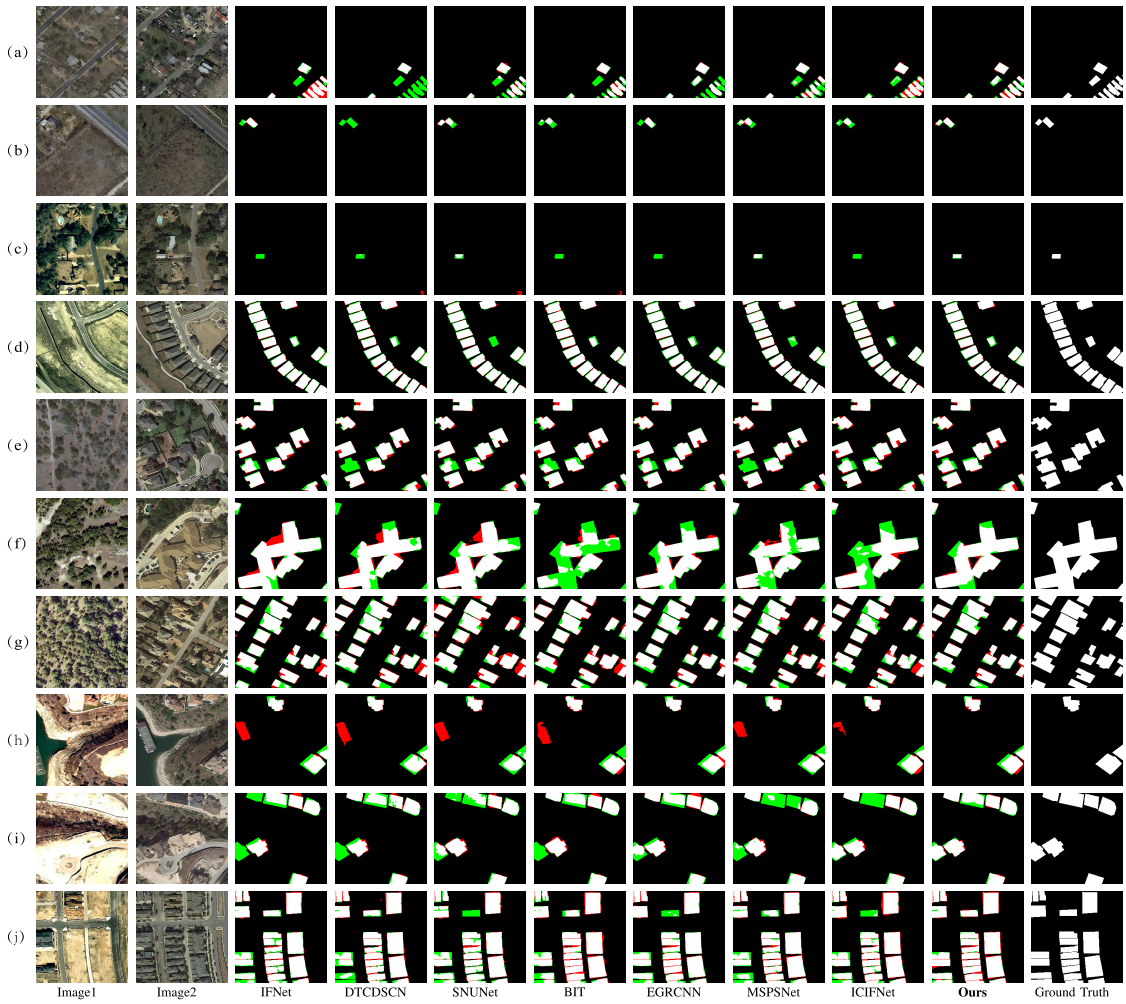


Fig. 6. Visual results of different models on LEVIR-CD. (a)–(j) Ten representative samples. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

C. Performance Comparison

Numerically, Tables II and III report the performance of all the competing methods on six metrics. Visually, Fig. 5 shows the F1 values and their standard deviation on all the four datasets and several selected competitors. Evidently, the results assert the superior behavior of our DMINet over others. For example, F1 values generated by our DMINet exceed the most recently published ICIFNet by a significant margin. Taking Kappa as an instance, the average improvements of our model over ICIFNet, MSPSNet, EGRCNN, BIT, and SNUNet are 1.0, 2.7, 2.2, 3.6, and 3.5, respectively. We believe these gains are remarkable since all the competitors are recently reported as the SOTA approaches. Qualitatively, Figs. 6–9 further illustrate the predicted maps on various datasets, where distinct colors are assigned to identify the correctness or incorrectness of the detection, including TP (white), TN (black), FP (red), and FN (green).

Visualization on LEVIR-CD (Fig. 6): Several representative samples are selected for visual comparison, such as extremely tiny objects scattered or densely distributed in Fig. 6(a)–(d), buildings with various sizes obscured by trees in Fig. 6(e)–(g), and exposure-affected buildings in Fig. 6(h) and (i). As illustrated in Fig. 6(a)–(d), for tiny buildings, the detection accuracy of our DMINet enjoys tremendous superiority over other

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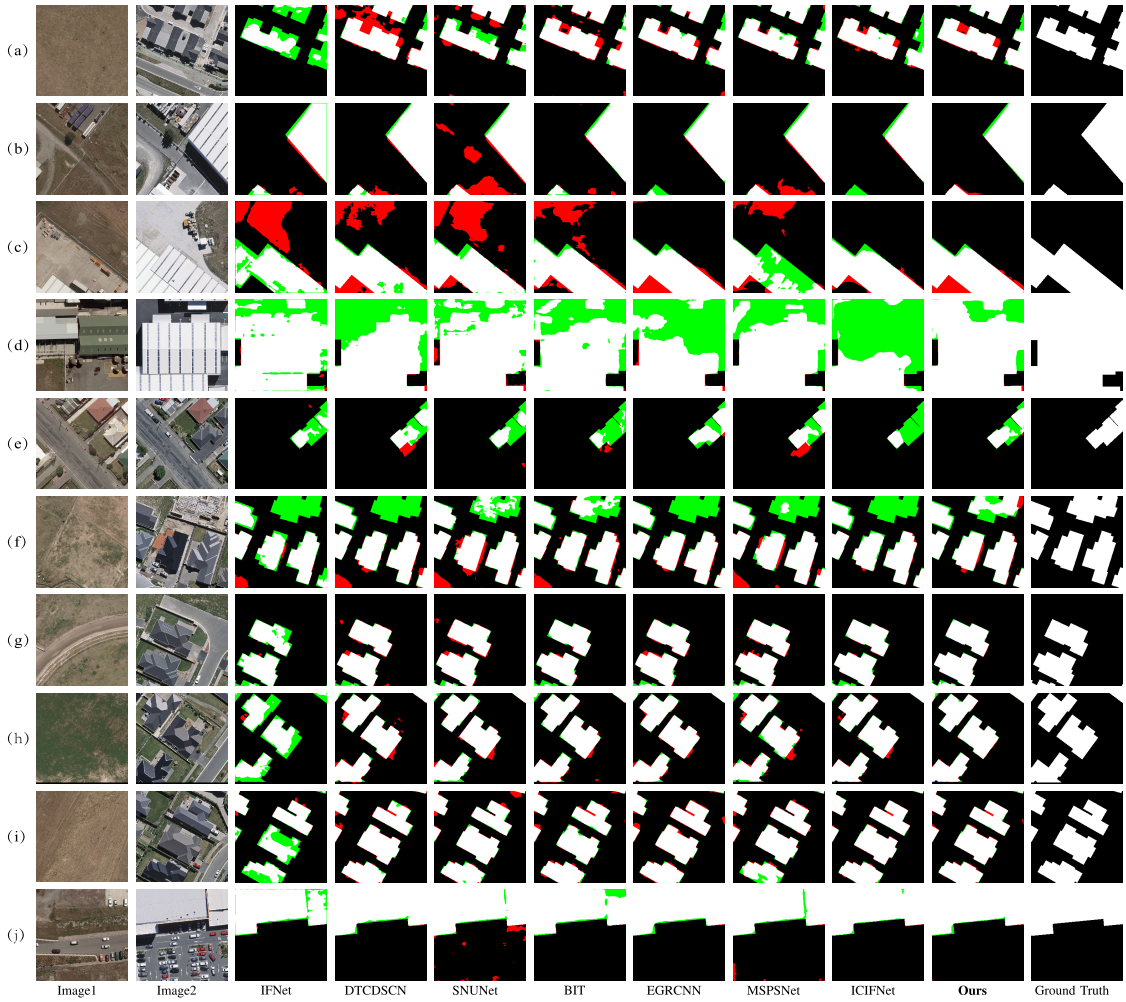


Fig. 7. Visual results of different models on WHU-CD. (a)–(j) Ten representative samples. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

competitors that suffer from severe omissions. This verifies the advantage of DMINet that revolves around the acquisition and aggregation of multilevel differences in full stages. For Fig. 6(f), fewer omission faults and commission errors authenticate the capability of our DMINet on detecting changes in scenes with sophisticated structures. Note in this scenario that the post-built constructions (located in the bottom-right corner of Image 2) increase the complexity of recognition due to their positioning in the overexposed regions of Image 1. Moreover, comparing the false identifications of each model in Fig. 6(h) and (i), our DMINet attains a surprisingly precise discriminability under occlusion and illumination interference. In a nutshell, the qualitative comparisons are consistent with the quantitative results in Table II, where our proposed DMINet reaches SOTA attainment on LEVIR-CD.

Visualization on WHU-CD (Fig. 7): Introducing the similar sampling scheme for the visual comparison, representative instances from WHU-CD are selected, including the serious interference from surfaces covered with cement or asphalt in Fig. 7(a)–(c), the visually distinct shed in Fig. 7(d), and tiny-sized targets or dense buildings in Fig. 7(e) and (i). Although the changed regions in Fig. 7(a)–(c) are practically accessible for detection, mitigating interference-induced miss-recognition is a challenging point, where our DMINet delivers low false alarms in the visual presentation relative to the other competitors. In Fig. 7(d)–(f), the outcomes of our DMINet contain noticeably lower omissions regarding the large shed and tiny objects. As for considerable normative buildings as illustrated in Fig. 7(g)–(i), our DMINet identifies them with greater comprehensiveness.

Visualization on GZ-CD (Fig. 8): Several image pairs from GZ-CD, which witnessed the construction process in the suburbs of Guangzhou, China, are used. Representatively, in Fig. 8(a)–(e), the unchanged pixels are vulnerable to misinterpretation since various exposure levels occur in the acquisition procedure of the bitemporal images. For both these subfigures and Fig. 8(f)–(j), distinguishing various types of changed buildings such as dense containers, residence houses, large-scale factories, and high-rise structures is fairly problematic. Encouragingly, albeit with extremely limited image pairs relative to other datasets, our DMINet enjoys the optimal visual behavior on GZ-CD, which confirms closely to the quantitative results in Table II.

Visualization on SYSU-CD (Fig. 9): The SYSU-CD dataset enjoys a broad range of target types, such as vegetations in Fig. 9(a) and (b), cultivated fields in Fig. 9(c) and (d),

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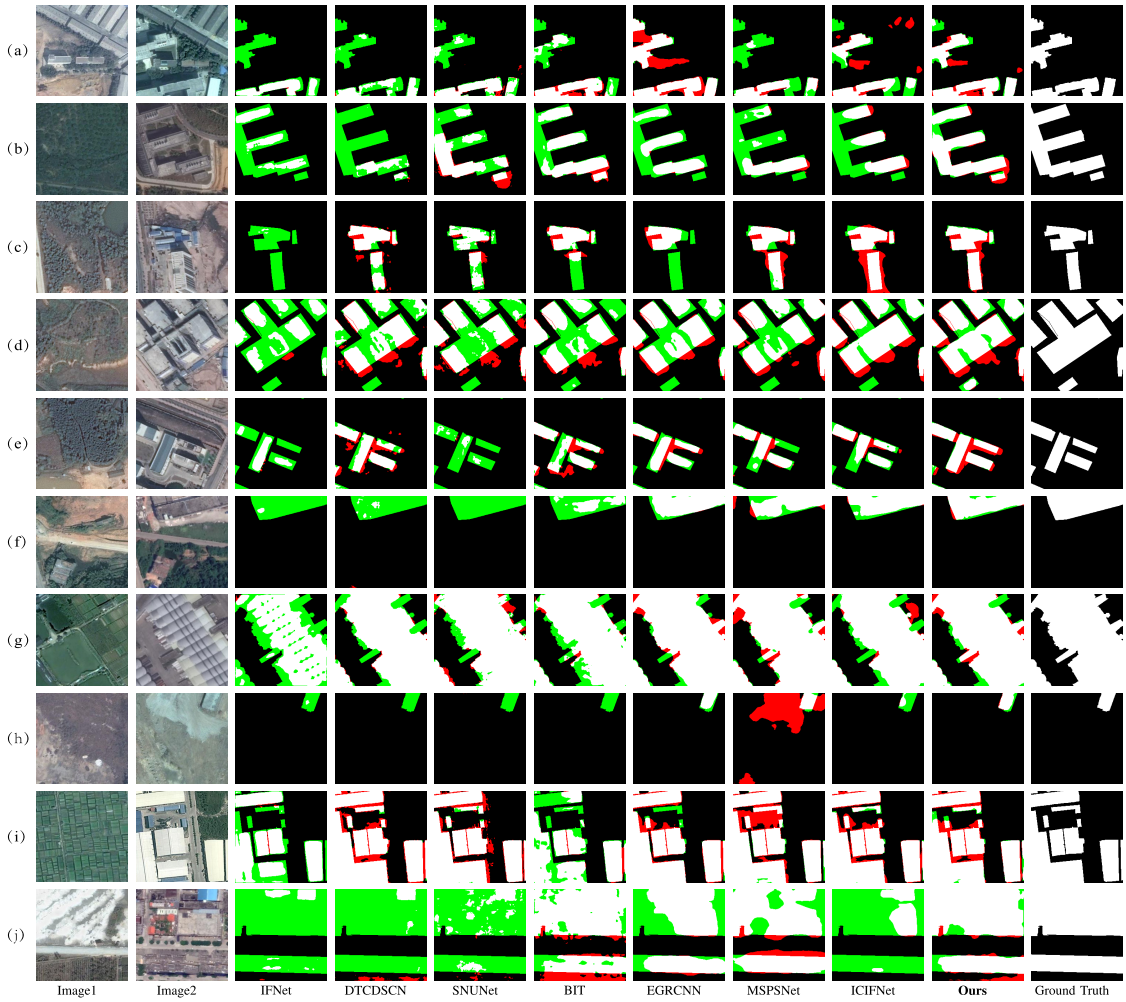


Fig. 8. Visual results of different models on GZ-CD. (a)–(j) Ten representative samples. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

pathways in Fig. 9(e) and (f), buildings in Fig. 9(g) and (h), and vessels in Fig. 9(i) and (j). Notwithstanding the challenges, our DMINet delivers exceptional visualization results that concord closely with those in Table III. For example, in Fig. 9(b), the results of our DMINet are evidently immune to building shadows. In Fig. 9(c) and (d), DMINet leaves a noticeable gap in the detection of cultivated fields. Moreover, in Fig. 9(i) and (j), DMINet is proficient in detecting vessels as well. In a nutshell, both quantitative and qualitative results indicate the serviceability and superiority of DMINet, which consistently yields higher positive rates and lower commission errors than the others.

To further evidence the efficiency of the proposed model, Table I presents the number of model parameters (Params.) and the floating-point operations per second (FLOPs), for which the input image pair is of size $256 \times 256 \times 3$. Comparatively, attributing to the elaborate yet lightweight architecture design, our DMINet outperforms ICIFNet in terms of both efficacy and efficiency. Although BIT and MSPSNet hold fewer parameters and FLOPs, the lead is marginal. Moreover, these two methods perform much poorer than DMINet in terms of efficacy.

To investigate the robustness and generalization of the models in cases of limited samples, we conduct experiments on 5%, 20%, and 50% training image pairs, as illustrated in Fig. 10. What is particularly striking is that DMINet constantly occupies or shares the first place as fluctuations in sample sizes occur. For instance, the detection scores of both MSPSNet and DMINet are decent on 5% samples of LEVIR-CD, WHU-CD, and GZ-CD, yet MSPSNet lags behind DMINet with a huge margin on SYSU-CD. Similarly, DMINet outperforms ICIFNet with superior generalization on LEVIR-CD and SYSU-CD. The presented results highlight the favorable effects that our proposed model delivers great adaptability and stability in practical applications with rarely available samples, which is certainly an accomplishment for DMINet with its inherently fast inference speed.

D. Ablation Analysis

Intended to further confirm the contributions of respective modules and various branches of the proposed model, as presented in Tables IV–VI, comprehensive ablation experiments are conducted in this section.

1) Methods of Acquiring Difference Maps: In the latest research on few-shot object detection [53], a conjunction of multiple options, including elementwise subtraction

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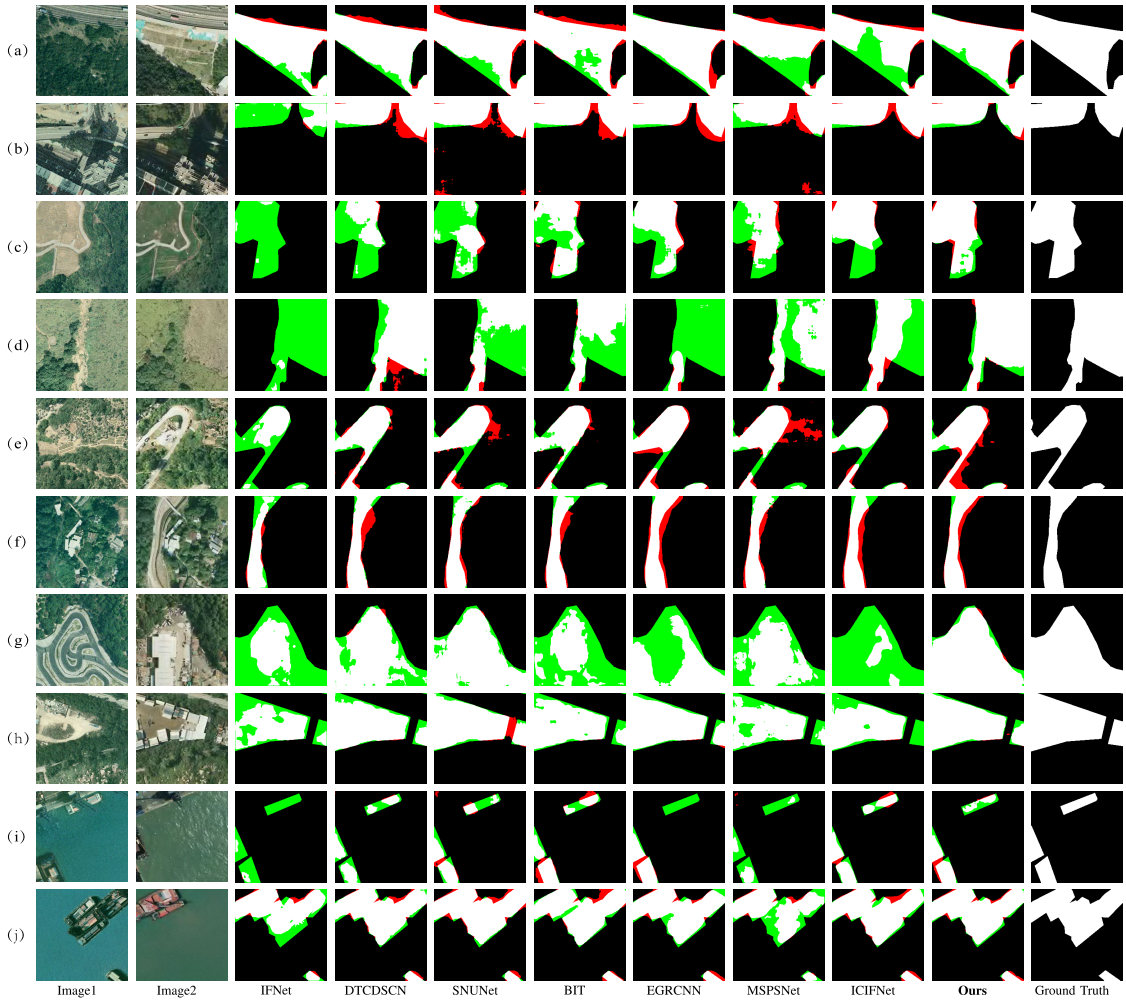


Fig. 9. Visual results of different models on SYSU-CD. (a)–(j) are ten representative samples. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

TABLE IV

ABLATION STUDY ON DIFFERENT METHODS OF ACQUIRING DIFFERENCE MAPS

Sub.	Mult.	Cat.	LEVIR-CD	WHU-CD	GZ-CD	SYSU-CD
✓	×	×	90.31	86.38	85.67	81.51
×	✓	×	90.20	86.86	85.79	80.32
×	×	✓	90.62	85.77	85.37	80.61
✓	✓	✓	90.17	86.43	85.17	81.06
×	✓	✓	90.47	87.84	84.94	80.88
✓	✓	×	88.39	86.01	83.55	79.28
✓	×	✓	**90.71**	**88.69**	**86.52**	**82.08**

TABLE V

ABLATION STUDY ON DIFFERENT ATTENTION MECHANISMS

Model	LEVIR-CD	WHU-CD	GZ-CD	SYSU-CD
Without	89.95	86.72	85.26	80.60
Self-attention	90.39	87.39	85.61	81.38
Cross-attention	90.43	88.04	85.83	81.17
Joint-attention	**90.71**	**88.69**	**86.52**	**82.08**

TABLE VI

ABLATION STUDY ON MULTILEVEL FEATURE ALIGNMENT

Multi-levels & Aggregation			LEVIR -CD	WHU -CD	GZ -CD	SYSU -CD
64×64	32×32	16×16	-	-	-	-
✓	×	×	88.95	83.45	80.69	77.25
×	✓	×	90.15	85.96	81.27	77.86
×	×	✓	88.55	85.97	83.33	79.73
×	✓	✓	89.95	87.34	**86.73**	79.89
✓	✓	×	90.45	86.93	84.01	79.04
✓	✓	✓	**90.71**	**88.69**	86.52	**82.08**
Only Concatenation			90.53	87.28	85.94	81.25
Ours			**90.71**	**88.69**	**86.52**	**82.08**

(Sub.) and multiplication (Mult.) calculations, and the learnable concatenation (Cat.) operation, accesses the similarity features between inputs. Accordingly, as demonstrated in Table IV, the ablation experiments of the three operations are performed to derive the difference features between the bitemporal RS images. This table is quite revealing that the results are decent on all the three datasets with a particular operation individually. However, it is optimal to adopt the two alternatives, i.e., Sub. and Cat., to capture the change maps.

2) Attention Mechanisms: The traditional dot-product Self-Att mechanism dynamically measures the similarities between

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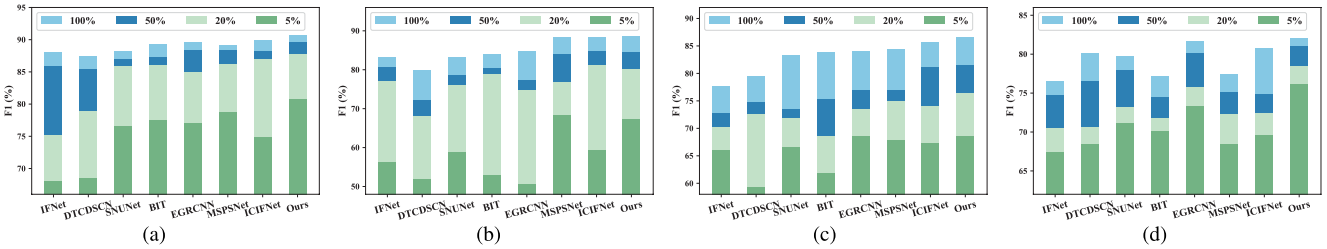


Fig. 10. Comparison on four datasets with different numbers of training samples, i.e., 5%, 20%, 50%, and 100%, where the bars indicate the F1 values. (a) Methods on LEVIR. (b) Methods on WHU. (c) Methods on GZ. (d) Methods on SYSU.

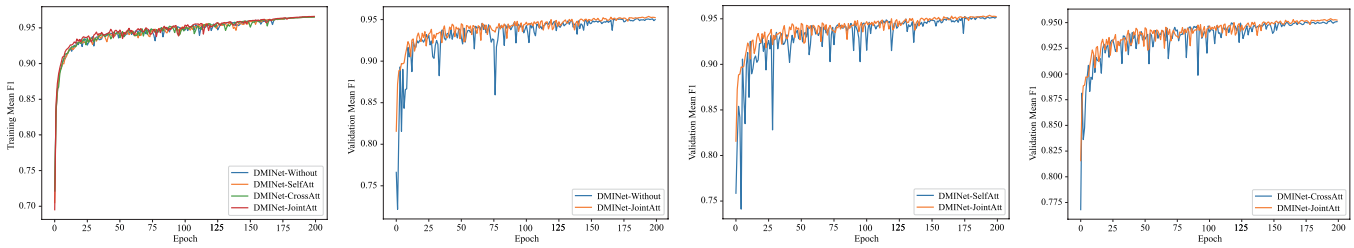


Fig. 11. Accuracy variations against the increasing training epoch. The mean F1-score is reported. (a) Training accuracy of DMINet with different attention mechanisms on the LEVIR-CD dataset. (b)–(d) Validation accuracy of DMINet with different attention mechanisms on the LEVIR-CD dataset.

TABLE VII
ABLATION STUDY ON COEFFICIENTS OF LOSS FUNCTION

λ_1=λ_2	λ_3=λ_4	LEVIR-CD	WHU-CD	GZ-CD	SYSU-CD
1	0	90.51	86.44	85.66	81.46
1	0.2	**90.75**	87.81	85.86	81.89
1	0.5	90.71	**88.69**	86.52	**82.08**
1	0.8	**90.75**	88.32	**86.79**	81.73
1	1	90.74	87.56	86.02	82.01

TABLE VIII
ABLATION STUDY ON DMINET WITH DIFFERENT BACKBONES

Backbones	Params. (M)	FLOPs (G)	LEVIR -CD	WHU -CD	GZ -CD	SYSU -CD
ResNet18	**6.24**	**14.42**	90.71	88.69	86.52	82.08
ResNet34	11.63	18.66	90.93	88.81	86.66	82.13
ResNet50	28.34	43.27	91.16	89.05	86.59	82.07
ResNet101	47.33	52.99	**91.29**	**89.12**	**86.75**	**82.24**

the representation vectors to enable the receptive field to cover the entire input space. In contrast, in the design of DMINet, we introduce an intertemporal JointAtt method, by sharing partial representations, for feature interaction between bitemporal RS images. Taking F1 metric as an example, Table V gives the ablation results on four datasets, while Fig. 11 illustrates the increasing accuracy against the epochs on LEVIR-CD. From Fig. 11(a), we can observe that different attention mechanisms share analogous performance on the training samples. However, as shown in Fig. 11(b)–(d), JointAtt enjoys a more stable curve on the validation set than the other attention mechanisms. The results on four datasets in Table V also confirm the same conclusion. The model using JointAtt leads the F1 values in all the cases, which demonstrates a better generalization performance for DMINet. In summary, we argue that applying JointAtt for the CD task benefits feature coupling of bitemporal images.

3) Multilevel Alignment: To probe the significance of interlevel feature alignment, Table VI gives the ablation results on four datasets. Note that the initial multilevel features in Fig. 1 are derived from the first three stages of ResNet18, whose spatial sizes are 64×64, 32×32, and 16×16, respectively. Evidently, the addition of interlevel differences incrementally leads to higher F1 values, demonstrating the favorable role of combining multiple-level features. Besides, the used alignment scheme further improves the detection performance, compared with the fusion using only concatenation. To sum up, the multilevel incremental aggregation contributes positively to CD performance.

4) Coefficients of Loss Function: To investigate the effectiveness of deep supervision on model optimization, as shown in Table VII, we ablate the tradeoff coefficients for various predictions in (7). First, since the predicted maps can be roughly divided into the semifinished ones $P_{\text{sub}}^{\text{sp}}$ and $P_{\text{cat}}^{\text{sp}}$, and the final ones P_{sub} and P_{cat}, we set $\lambda_1 = \lambda_2$ and $\lambda_3 = \lambda_4$ to treat each branch fairly. Second, with λ_1 and λ_2 being fixed at 1, we traverse the values of λ_3 and λ_4 within the range of [0, 1] to report the performance variations. From this table, we can see that the imposed deep supervision surely favors a better performance. However, numerically, the results are relatively stable against different settings of the tradeoff parameters. To ease the model selection, we suggest to set $\lambda_1 = \lambda_2 = 1$ and $\lambda_3 = \lambda_4 = 0.5$ in practice.

5) Effect Using Different Backbones: To verify the influence of various backbones on our model, we conduct ablation experiments on ResNet18, ResNet34, ResNet50, and ResNet101. The results are given in Table VIII. Evidently, deeper backbones deliver detection gains to some extent. However, the gains are achieved with the sacrifice of some more resource consumptions. In practice, we believe that ResNet18 is a

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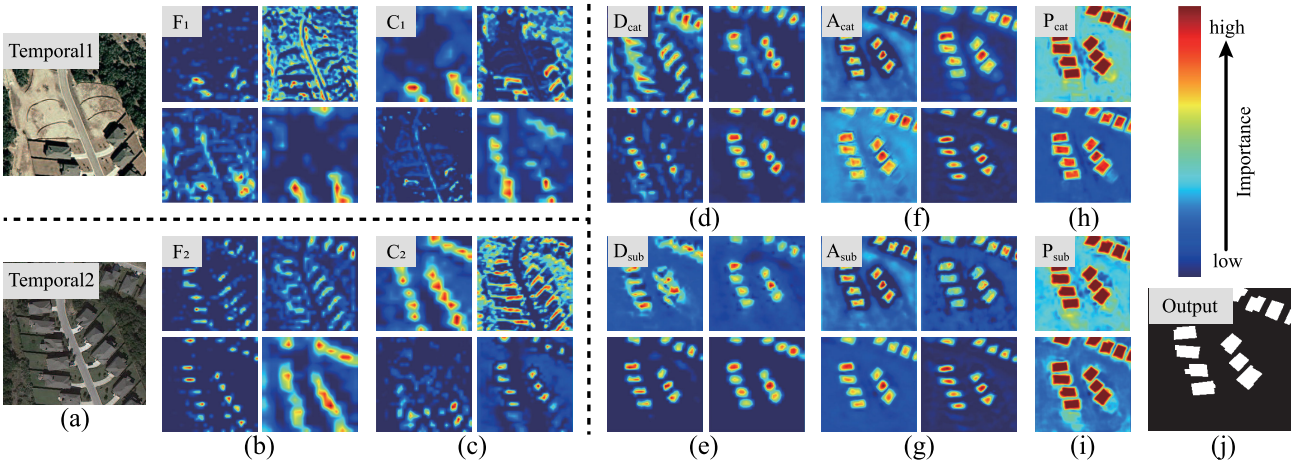


Fig. 12. Network visualization taking images from LEVIR-CD as examples. (a) Input images. (b) Selected multilevel feature maps F_1 and F_2 generated by CNN. (c) Selected feature maps C_1 and C_2 coupled through the intertemporal JointAtt module. (d) and (e) Selected difference features D_{cat} and D_{sub} obtained by concatenation and subtraction. (f) and (g) Selected feature maps $D_{\text{cat}}^{\text{agg}}$ and $D_{\text{sub}}^{\text{agg}}$ accessed by aggregating D_{cat} and D_{sub}, respectively, through the multilevel feature alignment module. (h) and (i) Change probability maps P_{cat} and P_{sub}. (j) Change map.

desirable choice with both satisfactory performances and saved resource consumption.

E. Network Visualization

Aiming at visualizing the feature maps at successive stages of our DMINet, we deliver an instance with images from LEVIR-CD. Through channelwise visualization, we select representative cases to be presented in Fig. 12. Given a pair of bitemporal inputs [Fig. 12(a)], the multilevel feature maps F_1 and F_2 [Fig. 12(b)] are first captured by Siamese CNN (ResNet18). Then, the information interaction between bitemporal F_1 and F_2 is performed by intertemporal JointAtt module, generating the coupled feature maps C_1 and C_2 [Fig. 12(c)]. Apparently, benefitting from the JointAtt mechanism by combining both self- and cross-attentions, the interaction process steers the attention of the model to changed regions and enforces the task-irrelevant interference to be suppressed. After that, the difference maps D_{cat} and D_{sub} [Fig. 12(d) and (e)] are accessed by two approaches, i.e., concatenation along the channels and elementwise subtraction, whose localization of targets and contours are somewhat coarse. Naturally, we introduce the multilevel difference aggregation module to progressively aggregate low-level detail information and high-level structural features, resulting in superior feature maps $D_{\text{cat}}^{\text{agg}}$ and $D_{\text{sub}}^{\text{agg}}$ [Fig. 12(f) and (g)]. The change probability maps P_{cat} and P_{sub} [Fig. 12(h) and (i)] after convolutions exhibit accurate target localization and refined boundaries, which demonstrates the effectiveness of the proposed modeling strategies.

V. Conclusion

To exploit the interactively spatial–temporal contextual information modeling between RS image pairs, in this article, we propose a DMINet for the CD task. Specifically, by incorporating SelfAtt and CrossAtt in a single module, DMINet leverages an intertemporal JointAtt to steer the attention distribution toward genuinely changed regions. In addition, the task-irrelevant variations between bitemporal images would be naturally suppressed. Afterward, to pay more attention on the difference features, the overall architecture of DMINet is conceived mainly from two perspectives, i.e., feature acquisition and multilevel aggregation. The former applies a dual branch of pixelwise subtraction and channelwise concatenation, while the latter relies on simple Hadamard product operations for incremental aggregation of the features. Extensive experiments on four publicly available datasets, especially in cases with limited training samples, demonstrate that our DMINet based on pretrained CNN (ResNet18) can achieve SOTA performance compared with other emerging competitors with both efficiency and effectiveness. Although designed for CD, the proposed DMINet method is exactly equipped to be transferred to other applications. By further leveraging the superiority of feature interaction between bitemporal images, in future research work, we intend to venture into CD on heterogenous RS image pairs captured by multisensors.

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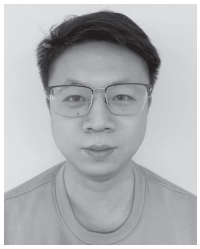
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